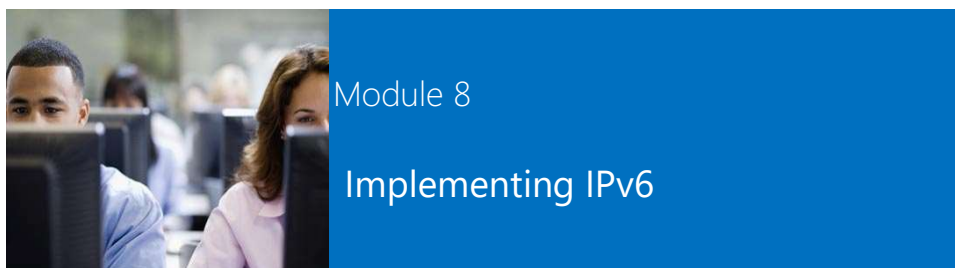


Microsoft® Official Course



Microsoft

Module Overview

- Overview of IPv6
- IPv6 Addressing
- Coexistence with IPv4
- IPv6 Transition Technologies

Lesson 1: Overview of IPv6

- Benefits of IPv6
- Differences Between IPv4 and IPv6
- IPv6 Address Format

Benefits of IPv6

Benefits of IPv6 include:

- Larger address space
- Hierarchical addressing and routing infrastructure
- Stateless and stateful address configuration
- Required support for IPsec
- End-to-end communication
- Required support for QoS
- Improved support for single-subnet environments
- Extensibility

Differences Between IPv4 and IPv6

Feature	IPv4	IPv6
Fragmentation	Performed by routers and sending host	Performed only by sending host
Address resolution	Broadcast ARP request frames	Multicast Neighbor Solicitation messages
Manage multicast group membership	IGMP	Multicast listener discovery
Router discovery	ICMP Router Discovery (optional)	ICMPv6 Router Solicitation and Router Advertisement (required)
DNS host records	A records	AAAA records
DNS reverse lookup zones	IN-ADDR.ARPA	IP6.ARPA
Minimum packet size	576 bytes	1280 bytes

IPv6 Address Format

- 128-bit address in binary:
00100000000000010000110110111000000000000000
00000010111100111011000000101010101000000000
111111111111110001010001001110001011010
- 128-bit address divided into 16-bit blocks:
0010000000000001 0000110110111000
0000000000000000 0010111100111011
0000001010101010 0000000011111111
1111111000101000 1001110001011010
- Each 16-bit block converted to hex (base 16):
2001:0DB8:0000:2F3B:02AA:00FF:FE28:9C5A
- Further simplified by removing leading zeros:
2001:DB8:0:2F3B:2AA:FF:FE28:9C5A



IPv6 Address Format

[0010][1111][0011][1011]



IPv6 Address Format

[0010][1111][0011][1011]

8 4 2 1
[0 0 1 0]
 $0+0+2+0=2$

[1 1 1 1]
 $8+4+2+1=F$

[0 0 1 1]
 $0+0+2+1=3$

[1 0 1 1]
 $8+0+2+1=B$



IPv6 Address Format

[0010][1111][0011][1011]

8 4 2 1
[0 0 1 0]
 $0+0+2+0=2$

[1 1 1 1]
 $8+4+2+1=F$

[0 0 1 1]
 $0+0+2+1=3$

[1 0 1 1]
 $8+0+2+1=B$

= 2F3B



IPv6 Address Format

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00100000000000010000110110111000000000000000
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Lesson 2: IPv6 Addressing

- IPv6 Address Structure
- Global Unicast Addresses
- Unique Local Unicast Addresses
- Link-Local Unicast Addresses
- Autoconfiguring IPv6 Addresses
- Demonstration: Configuring IPv6 Client Settings

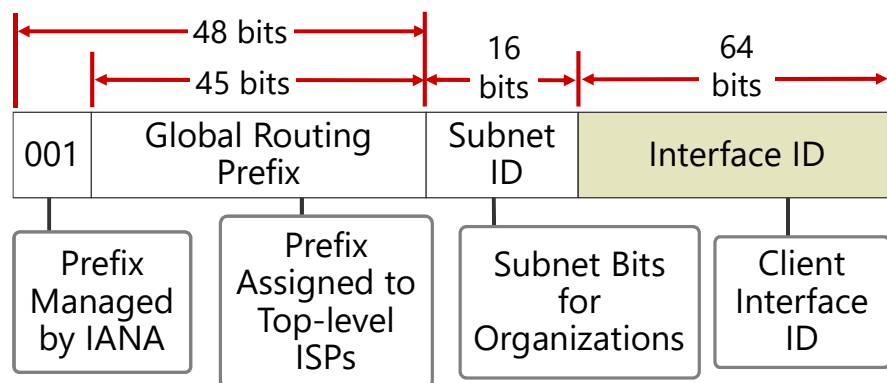
IPv6 Address Structure

- The number of network bits is defined by the prefix
- Each host has 64-bits allocated to the interface identifier

Type of address	IPv4 address	IPv6 address
Unspecified	0.0.0.0	::
Loopback	127.0.0.1	::1
Autoconfigured	169.254.0.0/16	FE80::/64
Broadcast	255.255.255.255	Uses multicasts instead
Multicast	224.0.0.0/4	FF00::/8

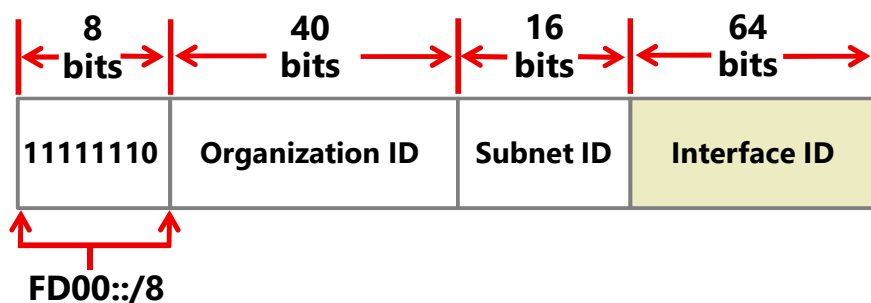
Global Unicast Addresses

- Are routable on the Ipv6 Internet
- Allocate 16 bits for internal subnetting
- Begin with 2 or 3 (2000::/3)



Unique Local Unicast Addresses

- Are equivalent to IPv4 private addresses
- Require the organization ID to be randomly generated
- Allocates 16 bits for internal subnetting

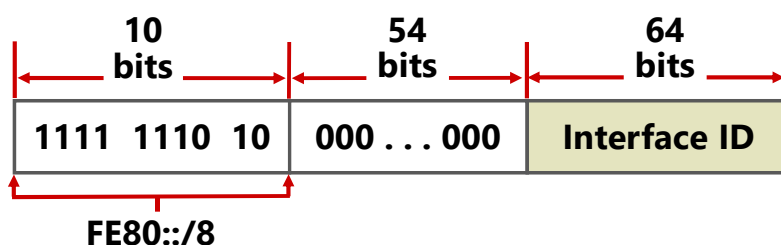


Link-Local Unicast Addresses

- Are automatically generated on all IPv6 hosts
- Are similar to IPv4 APIPA addresses
- Are sometimes used in place of broadcast messages
- Include a zone ID that identifies the interface

Examples: fe80::2b0:d0ff:fee9:4143%3

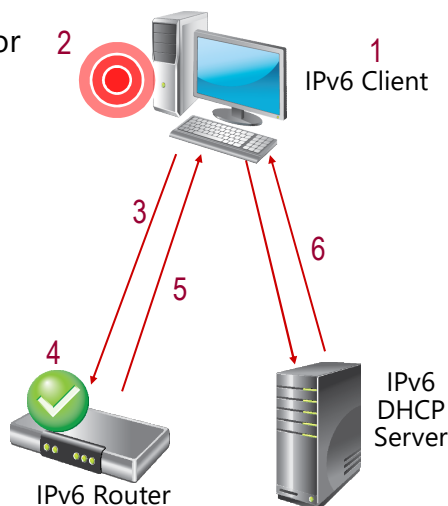
fe80::94bd:21cf:4080:e612%2



Autoconfiguring IPv6 Addresses

Address autoconfiguration for IPv6 is a six-step process:

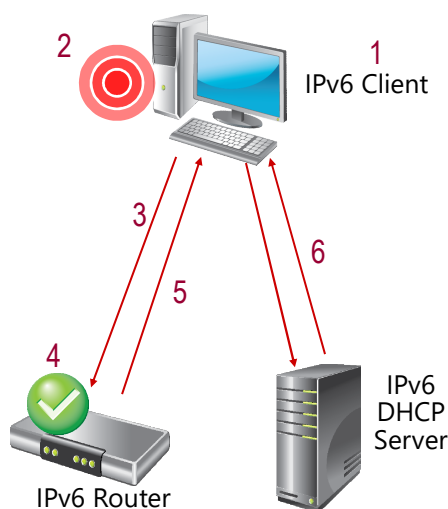
- 1 Derive a link-local address
- 2 Check for address conflicts by using neighbor solicitation
- 3 Check for a router on the network
- 4 Check the router for prefixes
- 5 Add prefixes
- 6 If Managed or Other flag set, check DHCPv6



Autoconfiguring IPv6 Addresses

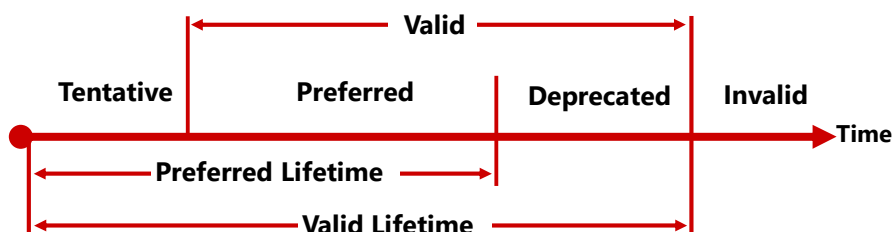
The six-step process:

- 1 Derive a link-local address
 - fe80::d593:e1e:e612:53e4%10
- 2 Check for address conflicts by using neighbor solicitation
- 3 Check for a router on the network
 - Router configuration search
- 4 Check the router for prefixes
- 5 Add prefixes
 - Additional router prefixes
- 6 If Managed or Other flag set, check DHCPv6
 - DHCPv6 information received



Autoconfiguring IPv6 Addresses

Autoconfigured IP Timeline



Demonstration: Configuring IPv6 Client Settings

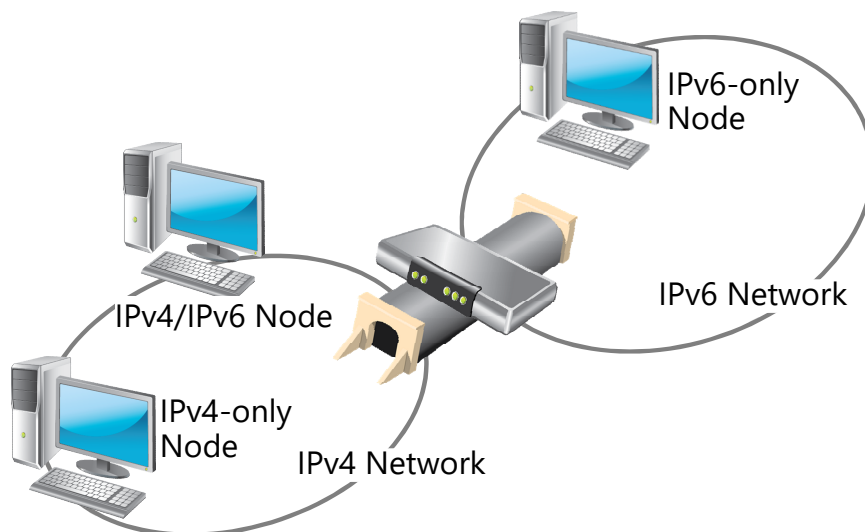
In this demonstration, you will see how to:

- View IPv6 configuration by using **ipconfig** and **Get-NetIPAddress**
- Configure IPv6 on a domain controller and a server
- Verify IPv6 communication is functional

Lesson 3: Coexistence with IPv4

- What Are Node Types?
- IPv4 and IPv6 Coexistence
- Demonstration: Configuring DNS to Support IPv6
- What Is IPv6 over IPv4 Tunneling?

What Are Node Types?



Coexistence entre IPv4 et IPv6

Windows Server 2012 utilise une architecture à double niveau : un niveau IPv4 et un niveau IPv6

A propos des enregistrement DNS :

- Hôte (A) : nœud IPv4
- Hôte (AAAA) : nœud IPv6
- Pointeur (PTR) : nœud IPv4 et nœud IPv6

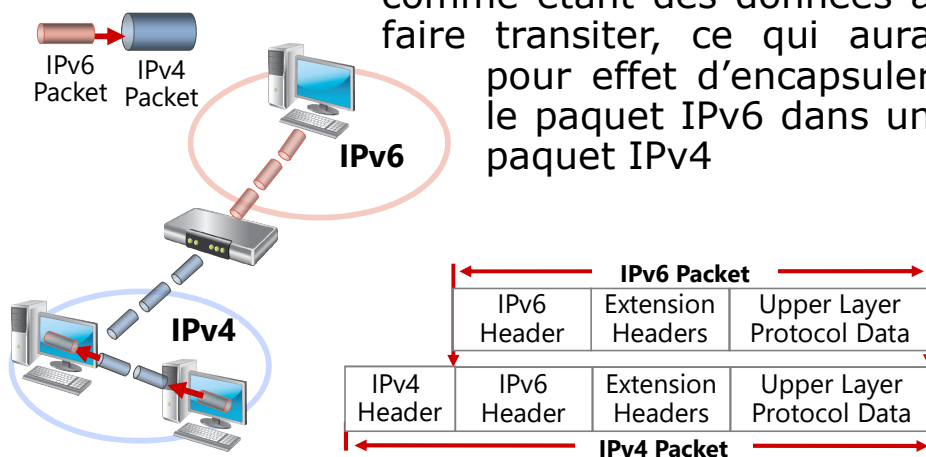
Demonstration: Configuring DNS to Support IPv6

In this demonstration, you will see how to:

- Configure an IPv6 host (AAAA) resource record for an IPv6 address
- Verify name resolution for an IPv6 host (AAAA) resource record

Coexistence IPv4-IPv6 : Tunneling

L'idée principale est, du point de vue du réseau IPv4, de considérer les paquets IPv6 comme étant des données à faire transiter, ce qui aura pour effet d'encapsuler le paquet IPv6 dans un paquet IPv4



Lesson 4: IPv6 Transition Technologies

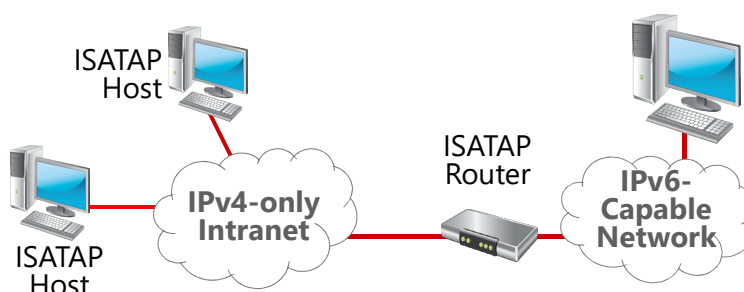
- What Is ISATAP?
- What Is 6to4?
- What Is Teredo?
- What Is PortProxy?
- Process for Transitioning to IPv6

ISATAP (Intra-Site Automatic Tunnel Address Protocol)

- ISATAP a été défini pour fournir une connectivité IPv6 à des équipements terminaux au sein de réseaux IPv4 et pour ainsi permettre un premier déploiement d'applications IPV6
- ISATAP utilise un format d'identificateur de machine qui inclut l'adresse IPv4 :

Adresse Privée: FD00::0:**5EFE**:192.168.137.133

Adresse Publique: 2001:db8::200:**5EFE**:131.107.137.133



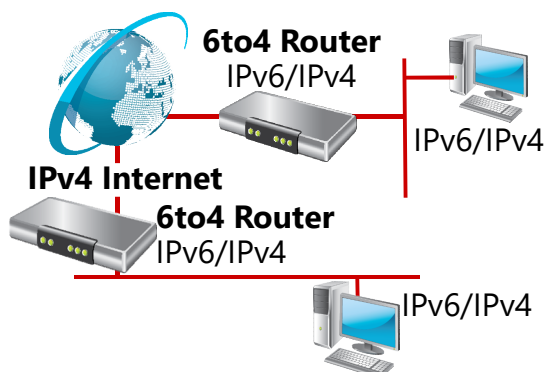
6to4

- La méthode 6to4 est utilisée pour relier des réseaux IPv6 à travers des réseaux IPv4
- Non compatible avec le NAT
- Le préfixe est construit à partir de l'adresse IPv4 :
2002:@IPv4:Subnet_ID::/64

Pour activer la fonctionnalité 6to4 :

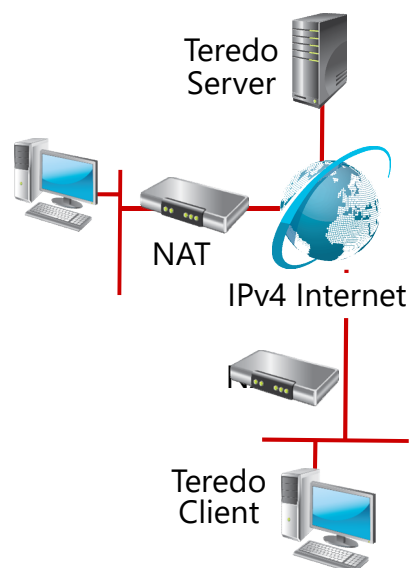
- Activer l'ICS (Internet Connection Sharing)
- Utiliser Windows PowerShell :

```
Set-Net6to4Configuration
```



Teredo

- La méthode Teredo est utilisée pour relier des réseaux IPv6 à travers des réseaux IPv4 lorsqu'un NAT se trouve dans la chaîne de communication
- Le protocole teredo requiert un serveur pour initialiser la communication
- Windows Power Shell
`Set-NetTeredoConfiguration`



Windows Server 2012:

- Peut-être configuré comme étant un client (par défaut), un serveur ou un relais

PortProxy (traduction)

Utiliser PortProxy :

- Pour fournir à des machines utilisant l'IPv6 uniquement l'accès aux applications développées pour tourner avec de l'IPv4 uniquement.
- Comme proxy entre les hôtes IPv4 uniquement et IPv6 uniquement

Limitations :

- Limité aux applications TCP
- Les informations d'adresse intégrées dans un paquet ne peuvent pas être modifiées

Process for Transitioning to IPv6

To transition from IPv4 to IPv6 you must:

- Update applications to support IPv6
- Update routing infrastructure to support IPv6
- Update devices to support IPv6
- Update DNS with records for IPv6
- Upgrade hosts to IPv4/IPv6 nodes

Lab: Implementing IPv6

- Exercise 1: Configuring an IPv6 Network
- Exercise 2: Configuring an ISATAP Router

Logon Information

Virtual machines	20410D-LON-DC1 20410D-LON-RTR 20410D-LON-SVR2
User name	Adatum\Administrator
Password	Pa\$\$w0rd

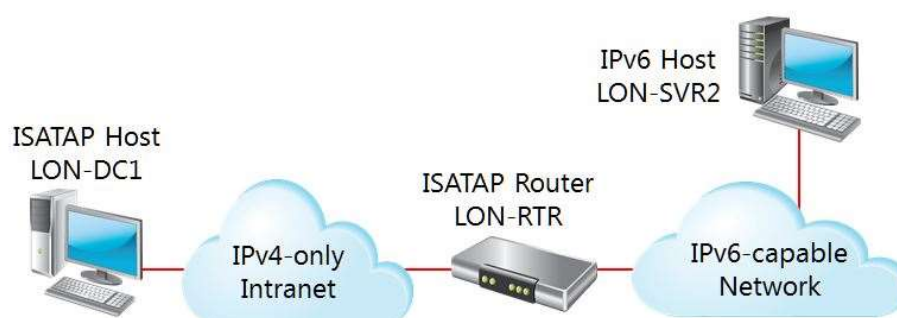
Estimated Time: 45 minutes

Lab Scenario

The IT manager at A. Datum Corporation has been briefed by several program and application vendors about newly added support for IPv6 in their products. A. Datum Corporation does not have IPv6 support currently. However, the IT manager wants you to configure a test lab that uses IPv6. As part of the test lab configuration, you also need to configure ISATAP to allow communication between an IPv4 network and an IPv6 network.

Lab Scenario

This is the layout of the completed test environment



Lab Review

- Did you configure IPv6 statically or dynamically in this lab?
- Why did you not need to configure LON-DC1 with the IPv4 address of the ISATAP router?

Module Review and Takeaways

- Review Questions
- Best Practices

